

Foundations Cheat Sheet

Networking, subnetting (binary), coding

BINARY OCTET BIT VALUES

128 64 32 16 8 4 2 1 (192 = 128+64 = 11000000)

CIDR MASKS & BLOCK SIZE (BLOCK = 256 - MASK)

/25 128 · /26 192 · /27 224 · /28 240 · /29 248 · /30 252

USABLE HOSTS ($2^{\text{HOST}} - 2$)

/24 254 · /25 126 · /26 62 · /27 30 · /28 14 · /29 6 · /30 2

KEY RANGES

Private 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16

APIPA 169.254.0.0/16 · Loopback 127.0.0.1 · IPv6 loopback ::1

Classes A 1-126 · B 128-191 · C 192-223

PORTS

20/21 FTP · 22 SSH · 23 Telnet · 25 SMTP · 53 DNS · 67/68 DHCP · 80 HTTP · 88 Kerberos

110 POP3 · 123 NTP · 143 IMAP · 161 SNMP · 389 LDAP · 443 HTTPS · 445 SMB · 3389 RDP

MODELS & PROTOCOLS

OSI Physical, Data Link, Network, Transport, Session, Presentation, Application

TCP handshake SYN, SYN-ACK, ACK · TCP reliable / UDP fast

Helpers DNS names->IP · DHCP DORA · NAT private<->public · ARP IP->MAC

PYTHON QUICK REFERENCE

def f(x): ... for x in list: if/elif/else try/except

format(n,'08b') len(list) list [] / dict {} / set() / tuple ()

Big-O index $O(1)$ · binary search $O(\log n)$ · linear $O(n)$ · nested $O(n^2)$

Security+ Cheat Sheet

SY0-701 · 90 min, up to 90 Qs, pass 750/900, valid 3 yrs

FOUNDATIONS

CIA Confidentiality, Integrity, Availability · AAA = Authn, Authz, Accounting

Data states At rest, in transit, in use

CRYPTOGRAPHY

Symmetric One shared key, fast: AES, ChaCha20, 3DES

Asymmetric Key pair: RSA, ECC, Diffie-Hellman (key exchange)

Hashing One-way integrity: SHA-256 (MD5/SHA-1 weak) · salt + key stretching (bcrypt, PBKDF2)

PKI CA issues certs · CRL/OCSP revocation · CSR request · TLS secures HTTPS

CONTROLS

Function Preventive, detective, corrective, deterrent, compensating, directive

Category Technical, managerial, operational, physical

ATTACKS

Phishing / smishing / vishing / whaling · on-path (MITM) · SQLi · XSS · privilege escalation

DDoS · replay · pass-the-hash · supply chain · watering hole · zero-day

IDENTITY & NETWORK DEFENSE

MFA factors Know, have, are, somewhere, do · TOTP, FIDO2/passkey

Tools SIEM, SOAR, EDR, DLP · IDS detect / IPS block · WAF, NGFW, DMZ, 802.1X

OPERATIONS & GOVERNANCE

IR lifecycle Prepare, detect, analyze, contain, eradicate, recover, lessons learned

Resilience Backups: full/incremental/differential · RTO (downtime) vs RPO (data loss)

Governance Risk = likelihood x impact · policies, third-party risk, compliance (GDPR)

RHCSA Cheat Sheet

EX200 (RHEL 10) · 2.5 hrs, hands-on, pass 210/300

USERS, GROUPS, PERMISSIONS

```
useradd -u 1500 -G devs bob   passwd bob   usermod -L bob
chmod 640 f   chown user:grp f   setfacl -m g:devs:rwX /data   getfacl /data
```

Numeric `r=4 w=2 x=1 · 755=rwx r-x r-x · 640=rw- r-- --- · umask 022`

Special bits `SUID 4xxx (as owner) · SGID 2xxx (group) · sticky 1xxx (/tmp)`

STORAGE & LVM

```
lvcreate -L 1G -n lvdata vgdata   mkfs.xfs /dev/vgdata/lvdata
echo '/dev/vgdata/lvdata /data xfs defaults 0 0' >> /etc/fstab   mount -a
Grow lvextend -L +1G ...   then xfs_growfs /data   · lsblk, df -h
```

SERVICES, BOOT, CRON

```
systemctl start|enable|status|reload|mask <svc>   journalctl -u <svc>
```

Targets `systemctl set-default multi-user.target (CLI) / graphical.target · crontab -e`

NETWORK & FIREWALL

```
nmcli ...   firewall-cmd --permanent --add-service=http   firewall-cmd --reload
ssh-keygen   ssh-copy-id user@host   ss -tulpn
```

SELINUX

```
getenforce   setenforce 0   restorecon -Rv /path   setsebool -P <bool> on
```

Ports/contexts `semanage port -a -t <type> -p tcp <port> · semmanage fcontext ...`

CONTAINERS & PACKAGES

```
podman run ...   podman images   dnf install <pkg>   dnf list installed
```

AWS Cheat Sheet

CCP (CLF-C02, pass 700) then SAA (SAA-C03, pass 720)

COMPUTE & STORAGE

Compute EC2 (VMs) · Lambda (serverless, 15-min max) · ECS/EKS/Fargate (containers)

Storage S3 (object, 11 nines) · EBS (block) · EFS (shared file) · Glacier (archive)

S3 Classes Standard/IA/One Zone/Glacier/Intelligent · versioning, lifecycle, SSE-KMS

DATABASE

Relational RDS / Aurora · Multi-AZ = failover, read replica = read scaling

Other DynamoDB (NoSQL) · ElastiCache (cache) · Redshift (warehouse) · Athena (SQL on S3)

NETWORKING & SECURITY

VPC Public subnet -> Internet Gateway · Private -> NAT Gateway · endpoints, peering, TGW

Firewalls Security Group = stateful, instance · NACL = stateless, subnet

IAM Users (long-term) vs Roles (temporary, preferred) · JSON policies · least privilege · global

Keys/edge KMS (keys) · CloudFront (CDN) · Route 53 (DNS) · WAF/Shield

INTEGRATION & OPS

Decouple SQS (queue) · SNS (pub/sub) · EventBridge · Step Functions

Observe CloudWatch (metrics/logs) · CloudTrail (API audit) · Trusted Advisor

EXAM ESSENTIALS

Pricing On-demand · Reserved/Savings Plans · Spot (interruptible, cheapest)

Pillars Ops Excellence, Security, Reliability, Performance, Cost, Sustainability

Shared resp. AWS secures the cloud; you secure what is in it (data, config, patching EC2 OS)

PMP Cheat Sheet

2026 ECO · domains + agile; needs 35 hrs + experience

STRUCTURE

Domains People ~42% · Process ~50% · Business Environment ~8% (about half agile/hybrid)

Process groups Initiating, Planning, Executing, Monitoring & Controlling, Closing

THE PMI MINDSET (ANSWERS MOST QUESTIONS)

Be a proactive servant leader · understand the root cause first · empower the team · collaborate

EARNED VALUE & FORMULAS

EVM SPI = EV/PV (schedule) · CPI = EV/AC (cost) · below 1 = behind/over

Estimate PERT = $(O + 4M + P)/6$ · Channels = $n(n-1)/2$

Reserves Contingency = known risks (in baseline) · Management = unknown (outside)

SCHEDULE & SCOPE

Scope WBS -> work packages · scope creep (uncontrolled) · gold plating (unrequested)

Schedule Critical path = longest · float = slack · crashing (add resources) vs fast tracking (overlap)

Dependencies FS, SS, FF, SF · lead vs lag

AGILE

Scrum Roles: PO, Scrum Master, Developers · Events: Sprint, Daily, Review, Retro

Artifacts Product Backlog, Sprint Backlog, Increment · velocity · Definition of Done

CONTRACTS & PEOPLE

Contracts Fixed price (seller risk) · cost-plus (buyer risk) · T&M

Teams Tuckman: forming, storming, norming, performing, adjourning · Theory X/Y

CISSP Cheat Sheet

8 domains · think like a risk manager

DOMAINS (8)

1 Security & Risk Mgmt · 2 Asset Security · 3 Architecture & Engineering · 4 Comm & Network
5 IAM · 6 Assessment & Testing · 7 Security Operations · 8 Software Development Security

RISK

Formulas Risk = Threat x Vulnerability · ALE = SLE x ARO · SLE = Asset Value x Exposure Factor

Responses Avoid, transfer (insurance), mitigate, accept · residual risk remains after controls

ACCESS CONTROL

Models DAC (owner) · MAC (labels) · RBAC (role) · ABAC (attributes)

Principles Least privilege, need to know, separation of duties, defense in depth

SECURITY & THREAT MODELS

Confidentiality Bell-LaPadula: no read up, no write down

Integrity Biba: no write up, no read down · Clark-Wilson: well-formed transactions

STRIDE Spoofing, Tampering, Repudiation, Info disclosure, DoS, Elevation

CRYPTO

Keys Symmetric $n(n-1)/2$ · use GCM not ECB · Kerberos (KDC/TGT) · certs signed by a CA

OPERATIONS & CONTINUITY

IR Prepare, detect, respond, mitigate, report, recover, remediate, lessons

BCP/DRP BIA -> MTD · RTO < MTD · RPO (data loss) · hot/warm/cold sites

Frameworks NIST CSF (Identify, Protect, Detect, Respond, Recover) · ISO 27001 · ethics: society first

AI Innovator Cheat Sheet

Building with Claude, Codex, and agents

CORE TERMS

Model LLM predicts tokens (~3/4 word) · context window = max tokens · temperature = randomness

Adaptation Prompting (instructions) · RAG (retrieved context) · fine-tuning (weights: SFT/DPO/RFT)

RAG

Pipeline Chunk -> embed -> store (vector DB) -> retrieve top-k -> stuff prompt -> generate

Vector DBs Pinecone, Qdrant, Weaviate · semantic search via cosine similarity · reranking

RAG Triad Faithfulness, Answer Relevance, Context Relevance (measure with RAGAS)

AGENTS

Definition LLM + tools + loop + memory · loop: perceive, plan, act, observe, repeat (ReAct)

MCP Model Context Protocol: standard to connect agents to tools/data · Agent Skills = SKILL.md

Orchestrate LangGraph, CrewAI · guardrails/hooks · human-in-the-loop

PROMPT ANATOMY

Role + task + examples + output format + delimited input (few-shot, chain-of-thought)

PRODUCTION / LLMOPS

Ship FastAPI + Docker on AWS · observability: Langfuse, LangSmith · sandbox: E2B, Modal

Evals Track accuracy, latency, cost on every change · LLM-as-judge · defend prompt injection

TOOLS & PORTFOLIO

Tools Claude Agent SDK (pip install claude-agent-sdk) · OpenAI Codex (GPT-5.5)

Ship 3-5 RAG assistant · FastAPI+Docker LLM API · multi-agent workflow · MCP assistant · instrumented

Target Roles & Priority Map

AI Business Architect / Evangelist + Forward Deployed Engineer

TWO ADOBE-STYLE TARGET ROLES (A SPECTRUM)

Architect / Evangelist Business architecture + AI adoption; light coding; BI dashboards; TOGAF, PL-300, AI-900, Prosci

Forward Deployed Eng. Software engineering + MLOps; strong Python/PyTorch; SQL + evals + observability; AWS + Docker + CI/CD

Shared core AI literacy, RAG, agents, evals, customer empathy, communication

A2A MEANS TWO DIFFERENT THINGS

Architect role A2A = Agent-to-Agent (Google protocol: agents coordinate)

FDE role A2A = application-to-application (REST/gRPC, webhooks, events)

REPRIORITIZED ORDER (ROLE-FIRST)

1 Applied AI & agents · 2 Enterprise AI tools + protocols (AI-900) · 3 Business architecture (TOGAF)

4 Data & BI (PL-300) · 5 AI adoption (ADKAR/Prosci) · 6 CAPM/PMP · 7 AWS CCP · 8 Python

Engineer lean adds Track F (MLOps) + AWS SAA, Docker, CI/CD. RHCSA/CISSP optional unless a role needs them.

PORTFOLIO BEATS CERTS (BUILD ALL OF THESE)

MCP agent · Copilot Studio chatbot · capability map + value stream · Power BI adoption dashboard

AI enablement plan · an AI-backed API with tests, an eval harness, tracing, and a dashboard

Business Architecture Cheat Sheet

TOGAF Standard 10th Edition · BIZBOK

CORE IDEA

Business architecture links strategy to capabilities, value streams, and processes: the what before the how.

CAPABILITY MAP

Definition A stable, structure-agnostic view of what the business does (not how)

Dimensions People, Process, Information, Resources

Why stable Ignores org charts, which change often

VALUE STREAM

Definition End-to-end stages that deliver value to a stakeholder

Mapping Map capabilities to the stages that need them; heat-map for investment

TOGAF ADM

Method Phases A-H; Phase B = Business Architecture

Flow Current-state assessment -> target architecture -> gap analysis -> roadmap

MODELING & TOOLS

Diagrams ArchiMate (architecture) · BPMN (processes) · draw.io / Archi (free)

Body of knowledge BIZBOK for capability and value-stream practice

CERT

Target TOGAF Enterprise Architecture Foundation, then Practitioner

Prove it A capability map + a value stream for a real or sample business

Data Analytics & BI Cheat Sheet

Power BI (PL-300), Tableau, Excel

THE BI PIPELINE

Extract/Transform/Load (Power Query) -> model (star schema) -> measures (DAX) -> visualize -> story

MODELING

Star schema Fact table (events/measures) linked to dimension tables (attributes)

Measure A calculated aggregate, e.g. total sales

DAX context Row context (current row) vs filter context (applied filters)

VISUALS

Trend over time Line chart

Comparison Bar chart · Parts of a whole: pie (use sparingly)

Interactivity Slicers filter the report; a dashboard = one view of key KPIs

EXCEL & TABLEAU

Excel PivotTables + Power Query for cleaning and aggregation

Tableau Alternative viz tool; Desktop Specialist cert optional

ADOPTION METRICS (FOR THE EVANGELIST ROLE)

Track Active users, usage rate, time saved, ROI

Rule Prefer actionable metrics over vanity metrics

CERT

Target Microsoft PL-300 (Power BI Data Analyst)

Prove it An AI-adoption dashboard built in Power BI

AI Adoption & Change Cheat Sheet

Evangelism, ADKAR, metrics

ADKAR (PEOPLE ADOPT IN THIS ORDER)

A Awareness of the need to change

D Desire to support the change

K Knowledge of how to change

A Ability to apply the new skills

R Reinforcement to sustain it

Diagnose a stuck rollout by finding the first missing letter.

RUN ADOPTION LIKE A PRODUCT

Steps Pilot to prove value -> recruit AI champions -> workshops -> Center of Excellence

Resistance Address the why, not just the how

MEASURE & REPORT

Metrics Adoption rate, monthly active users, time saved, ROI

Exec reporting Impact, progress, and the next ask, kept simple

Lead vs lag Leading predicts; lagging confirms

EVANGELIZE

Build in public Publish what you build; present to varied audiences

Audience range VP-level to individual contributor

CREDENTIAL

Valued Prosci Change Management Certification (ADKAR is free to study)

Prove it An AI enablement plan + a workshop outline + an adoption dashboard

Enterprise AI Tools & Agent Protocols

Copilot, Glean, Cursor, Replit · MCP, A2A

ENTERPRISE AI TOOLS

Copilot / Copilot Studio M365 assistant; Copilot Studio builds low-code grounded agents

Glean Enterprise AI search/assistant over company data

Cursor / Replit AI code editor; browser IDE with an AI build agent

Adobe Firefly (images), Express, GenStudio (content supply chain)

AGENT PROTOCOLS (THE 2026 STACK)

MCP Model Context Protocol (Anthropic): agent -> tools and data

A2A Agent-to-Agent (Google): agents discover and delegate

Two-layer stack MCP vertical (tools) + A2A horizontal (agents)

Also ACP/UCP (agent commerce); OAP in the open-protocol landscape

DATA STRUCTURES FOR AI

Vector database Embeddings for similarity search (Pinecone, Qdrant, Weaviate)

Knowledge graph Entities + relationships (Neo4j); reasoning over connected data

CERT & PROOF

Cert Microsoft AI-900 (Azure AI Fundamentals)

Prove it A grounded Copilot Studio chatbot + an MCP-connected agent

AI Engineering & MLOps Cheat Sheet

Forward Deployed Engineer path

BUILD

Language / frameworks Python + PyTorch (tensors, training loop); FastAPI for services

Dev tooling Git, CI/CD, Docker; tests on everything

INTEGRATION (APPLICATION-TO-APPLICATION)

APIs REST (JSON/HTTP) and gRPC (binary/HTTP2)

Async Webhooks, queues, event-driven systems

Reliability Auth, rate limits, retries with backoff, idempotency

EVALUATION

Offline Golden test sets + acceptance thresholds

Online A/B tests, user rating flows, guardrails

Data work SQL + notebooks; segment metrics by customer/cohort/use case

OBSERVABILITY & LLMOps

Three pillars Logs, metrics, traces (OpenTelemetry)

Stack Prometheus + Grafana, Datadog; Langfuse/LangSmith for LLM traces

Safe rollout Feature flags + canary releases; watch for model drift

LLMOps Prompt/version management, eval harnesses, cost tracking

PROOF

Portfolio An AI-backed API with tests, an eval harness, tracing, and a dashboard